

Bilkent University, Department of Industrial Engineering
IE 468 - Pricing and Revenue Optimization
Lecture Notes #6

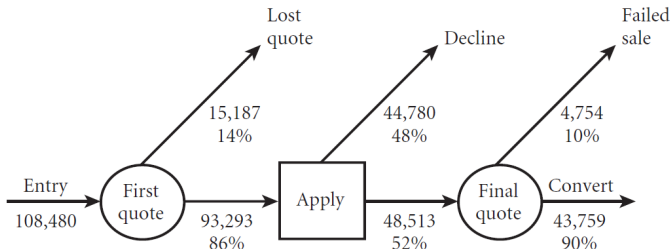
Alper Şen

Bilkent University

Fall 2025-2026

Customized Pricing

- List pricing: Customers arrive, observe price and decide whether or not to purchase
- Customized pricing: Potential customers approach the seller one by one, and the seller quotes each one a price
- Example: Telecommunication: selling services to corporate customers
- Example: 3PL services, package services, fleet sales
- Example: Consumer loans



Customized Pricing

- ① The customer signals his/her willingness to pay
 - ② The seller can quote a different price to each customer
 - ③ The seller can track lost business
- B2B setting: Bid or price quote, response to RFQ (Request for Quotation), or RFP (Request for Proposals)
 - B2C setting: Fill out an application form, phone call
 - List pricing and customized pricing may co-exist (MSRP as ceiling)

Bid-response curve

A seller who has been asked to bid on a potential deal.

Deal: A particular piece of business, a single order or a contract

Bid: Price we offer

Expected contribution at price p =

(Deal contribution at p) $\times$ (Probability of winning the bid at p)

Deal could be a complex bundle of products and/or services.

Why probability?

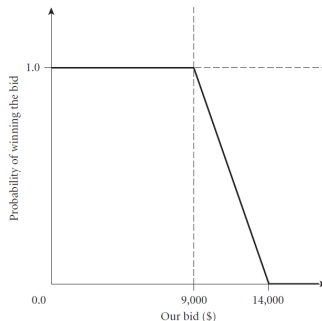
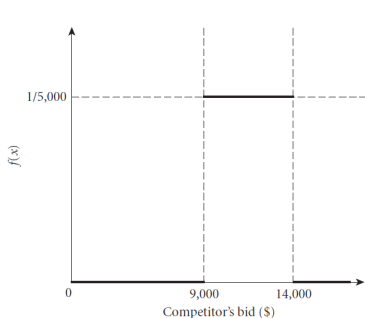
- Competitive uncertainty
- Preference uncertainty

Preference uncertainty may be absent

- Government: lowest bid, detailed specification of other terms but price
- Supplier auctions: initial selection of suppliers (vendor approval/certification)

Single Competitor Model

We are asked to bid for a deal. The customer is asking for 50 business computers with certain specifications. The computer costs as 10,000 TL per unit. We know that our competitor will have a bid between 9,000 TL and 14,000 TL (Uniformly distributed between 9,000 TL and 14,000 TL).



Single Competitor Model

f : Probability distribution function of the competitor's bid. (cdf F)

Bid-response function

$$\rho(p) = 1 - F(p) = \begin{cases} 1, & \text{if } p \leq 9,000 \\ 2.8 - p/5,000, & \text{if } 9,000 \leq p \leq 14,000 \\ 0, & \text{if } p \geq 14,000 \end{cases}$$

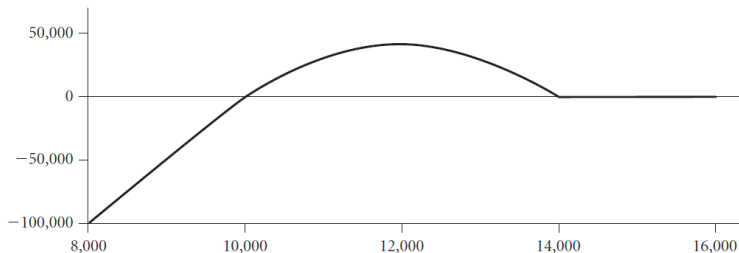
Expected profit

$$\tau(p) = \rho(p)m(p)$$

where $m(p)$ is the contribution at bid price p

$$\tau(p) = \rho(p) (p - 10,000) 50$$

Single Competitor Model



$$\max_p \tau(p) = (2.8 - p/5,000)(p - 10,000)50$$

$$\tau'(p^*) = -50(1/5,000)(p^* - 10,000) + 50(2.8 - p^*/5,000) = 0$$

$$p^* = 12,000$$

Probability of winning = $\rho(12,000) = 2.8 - 12,000/5,000 = 0.4 = 40\%$.

Expected profit = $(12,000 - 10,000) \times 50 \times 0.4 = 40,000$ TL

General Problem

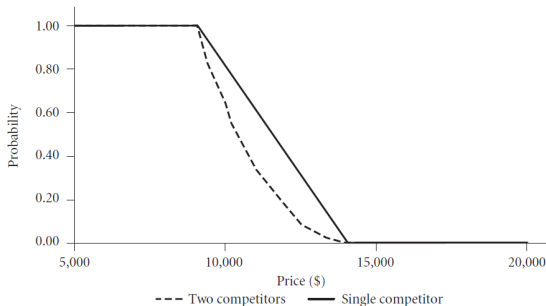
$$\max_p \tau(p) = \rho(p)m(p) = \rho(p)(p - c)d$$

$$\tau'(p^*) = d\rho'(p^*)(p^* - c) + d\rho(p^*) = 0$$

$$\boxed{-\frac{\rho'(p^*)p^*}{\rho(p^*)} = \frac{p^*}{p^* - c}}$$

Bid-response elasticity at the optimal bid price is equal to the inverse of the profit margin ratio at that price.

Multiple Competitors



Two identical competitors

$$\rho(p) = (1 - F(p))^2$$

In general

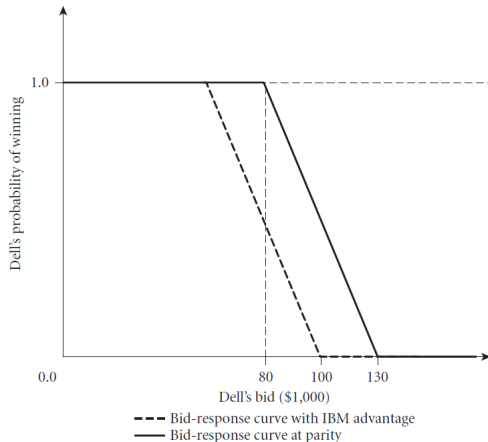
$$\rho(p) = \prod_{i=1}^n (1 - F_i(p))$$

Other Factors for Bid Selection

- Previous experience with supplier 10%
- Quality of proposal 5%
- Technical fit of solution 25%
- Price 60%

Bid Premium

Example: IBM's will be uniform between 80,000 and 130,000.
The customer believes that IBM's product is more valuable than Dell's.
(bid premium = 30,000)



Shifted bid-response curve for Dell when IBM enjoys a premium.

Bid-response Estimation

- Bottom-up modeling: Analysis of competitors
- Expert judgment
- Statistical analysis

Bid	Result
\$192.00	0
\$120.00	0
\$144.00	1
\$109.00	1
\$108.00	1
\$192.00	0
\$146.00	1
\$182.00	0
...	...
...	...
\$143.00	1
\$160.00	0
\$183.00	0

Maximum Likelihood Estimation

Logit bid-response

$$\rho(p) = \frac{1}{1 + e^{a+bp}}$$

p_i : price offered to i th customer

W_i : 1 if we won the bid, 0 otherwise

$$L_i(a, b) = \rho(p_i)W_i + (1 - \rho(p_i))(1 - W_i)$$

Likelihood and Log-Likelihood

$$L(a, b) = \prod_{i=1}^n L_i(a, b) \quad \ln(L(a, b)) = \sum_{i=1}^n \ln(L_i(a, b))$$

Maximum Likelihood Estimation

$$\max_{a, b} \sum_{i=1}^n \ln(L_i(a, b))$$